where $I = I_c + I_d$ which means the total current enclosed by the surface is sum of conduction current and displacement current. When a constant current is applied, displacement current $I_d = 0$ and hence $I_c = I$. Between the plates, the conduction current $I_c = 0$ and hence $I_d = I$.

EXAMPLE 5.1

Consider a parallel plate capacitor which is maintained at potential of 200 V. If the separation distance between the plates of the capacitor and area of the plates are 1 mm and 20 cm^2. Calculate the displacement current for the time in μs.

Solution

Potential difference between the plates of the capacitor, V = 200 V

The distance between the plates,

$$d = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$$

Area of the plates of the capacitor,

$$A = 20 \text{ cm}^2 = 20 \times 10^{-4} \text{ m}^2$$

Time is given in micro-second, $\mu s = 10^{-6}$ s

Displacement current

$$I_d = \varepsilon_\circ \frac{d\Phi_B}{dt} \Rightarrow I_d = \varepsilon_\circ \frac{EA}{t}$$

But electric field, $E = \frac{V}{d}$

Therefore,

$$I = \frac{V}{d} I_d = \varepsilon_\circ \frac{VA}{t\,d} = 8.85 \times 10^{-12} \times \frac{200 \times 20 \times 10^{-4}}{10^{-6} \times 1 \times 10^{-3}}$$

$$= 35400 \times 10^{-7} = 3.5\ mA$$

5.1.3 Maxwell's equations in integral form

Electrodynamics can be summarized into four basic equations, known as Maxwell's equations. These equations are analogous to Newton's equations in mechanics. Maxwell's equations completely explain the behaviour of charges, currents and properties of electric and magnetic fields. These equations can be written in integral form (or integration form) or derivative form (or differentiation form). The differential form of Maxwell's equation is beyond higher secondary level because we need to learn additional mathematical operations like curl of vector fields and divergence of vector fields. So we focus here only in integral form of Maxwell's equations:

1. First equation is nothing but the Gauss's law. It relates the net electric flux to net electric charge enclosed in a surface. Mathematically, it is expressed as

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enclosed}}{\varepsilon_\circ} \quad \text{(Gauss's law)} \quad (5.7)$$

where $\vec{E}$ is the electric field and $Q_{enclosed}$ is the charge enclosed. This equation is true for both discrete or continuous distribution of charges. It also indicates that the electric field lines start from positive charge and terminate at negative charge. This implies that the electric field lines do not form a continuous closed path. In other words, it means that isolated positive charge or negative charge can exist.

2. Second equation has no name. But this law is similar to Gauss's law in electrostatics. So this law can also be called as Gauss's law in magnetism. The

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